

Introduction

The Cochran-Mantel-Haenszel (CMH) test combines 2x2 tables across levels of a stratifying variable, producing a pooled odds ratio and a single test of association adjusted for the stratifier. It is the categorical analogue of Mantel-Haenszel stratified analysis in epidemiology and of two-way ANOVA without interaction.

Prerequisites

Fisher's exact test, chi-squared independence, odds ratios.

Theory

Given K 2x2 tables from K strata, the CMH statistic is

$$\chi^2_{\text{CMH}} = \frac{[\sum_k (a_k - E[a_k])]^2}{\sum_k \text{Var}(a_k)} \sim \chi^2_1 \text{ under } H_0.$$

Under the null of no association in any stratum, expected cell counts and variances are computed stratum by stratum.

The Mantel-Haenszel pooled odds ratio is

$$\widehat{OR}_{MH} = \frac{\sum_k a_k d_k / n_k}{\sum_k b_k c_k / n_k}.$$

This weights strata so that smaller strata contribute less; it is a weighted average of stratum-specific ORs.

The **Breslow-Day test** (or Woolf test) checks the assumption of homogeneity of the OR across strata. If homogeneity is rejected, the pooled OR is misleading.

Assumptions

- Independent observations within strata.
- Stratified 2x2 data.
- Homogeneous OR across strata (for the pooled OR to be meaningful).

R Implementation

```
# Three strata: age groups
# Each stratum: 2x2 table of exposure vs. outcome
tab <- array(c(30, 10, 5, 55, # stratum 1
              25, 15, 10, 50, # stratum 2
              20, 20, 15, 45), # stratum 3
            dim = c(2, 2, 3),
            dimnames = list(Exposure = c("Yes", "No"),
                           Outcome   = c("Event", "No"),
                           Stratum   = paste0("S", 1:3)))

mantelhaen.test(tab, correct = FALSE)

# Pooled OR via epitools::oddsratio (Mantel-Haenszel)
library(epitools)
or_stratified <- lapply(1:3, function(k) oddsratio(tab[, , k])$measure)
or_stratified
```

```
# Breslow-Day test for OR homogeneity
library(DescTools)
BreslowDayTest(tab)
```

Output & Results

Mantel-Haenszel chi-squared test without continuity correction

```
data: tab
Mantel-Haenszel X-squared = 67.5, df = 1, p-value < 2.2e-16
alternative hypothesis: true common odds ratio is not equal to 1
95 percent confidence interval: 4.82 13.87
sample estimates: common odds ratio 8.17
```

Breslow-Day Test on Homogeneity of Odds Ratios

X-squared = 1.5, df = 2, p-value = 0.47

Pooled OR = 8.17 (95 % CI 4.82 to 13.87); no evidence that stratum-specific ORs differ (Breslow-Day $p = 0.47$), so the pooled OR is a valid summary.

Interpretation

Reporting: “In a Mantel-Haenszel stratified analysis across three age groups, the pooled odds ratio was 8.17 (95 % CI 4.82 to 13.87; $\chi^2_{MH}(1) = 67.5$, $p < 0.001$). The Breslow-Day test gave no evidence of heterogeneity of association across strata ($p = 0.47$).”

Practical Tips

- Always test OR homogeneity (Breslow-Day) before reporting a pooled OR.
- For heterogeneous strata, report stratum-specific ORs or model the interaction.
- Logistic regression with the stratifier as a covariate is a more flexible modern alternative.
- CMH also has a conditional logistic version (`survival::clogit`) for individually matched pairs.
- Small strata may dominate the pooled estimate under MH weighting; inspect weights.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation

- Non-parametric tests